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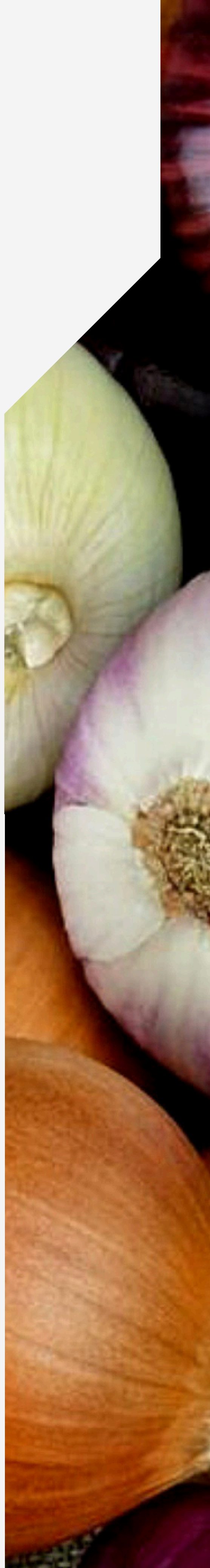

Analysis of the Seasonality of Onion Market Prices in the Philippines: Insights and Policy Recommendations from the Recent Price Volatility

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Abstract

The onion industry in the Philippines is a critical component of the agricultural sector, influencing both local economies and national food security. However, the market for onions, particularly red onions, has been affected by significant price volatility, primarily due to its seasonal production patterns and the imbalance between supply and demand. This study analyzes the seasonal fluctuations in the retail prices of red onions in the Philippines using time series data from 2012 to 2024. The analysis identifies a recurring trend in which prices are lowest from March to June, corresponding with the end of the harvest season, and highest from December to January, coinciding with the holiday season and the beginning of the next harvest cycle. These price fluctuations are further exacerbated by irregular price movements, such as those seen in 2022 and 2023, which were linked to supply shortages and policy interventions. Despite increased production in key onion-producing regions, such as Central Luzon and MIMAROPA, the Philippine onion market remains vulnerable to external shocks and inefficiencies within the supply chain. The findings highlight the need for targeted interventions, including the strengthening of cold storage and post-harvest infrastructure, improved supply chain transparency, and the development of region-specific policies that align importation strategies with domestic production cycles.

Keywords: onion, price volatility, seasonality, market fluctuations, Philippines

Introduction

The onion industry plays a crucial role in global agriculture, providing economic support to farmers and food manufacturers while serving as a staple ingredient with recognized health benefits (Griffiths et al., 2002). In the Philippines, onions are a key agricultural commodity, sustaining the livelihoods of many farmers and contributing to national food security. The country primarily cultivates two types of onions: bulb onions (red and yellow) and shallots, with bulb onions accounting for 79% of the total production area (DA-BAR, 2022). Despite advancements in cultivation techniques and technology (Domingo, 2023), the Philippine onion market experiences significant price volatility due to its seasonal production, which is largely concentrated in the dry season and limited to a few regions in Luzon, while demand remains steady throughout the year (DA-BAR, 2022). This seasonal imbalance contributes to fluctuating market prices, making locally produced onions more expensive than those from top onion-producing countries, where prices range from US$0.11/kg to US$0.43/kg, compared to the Philippine average of US$0.79/kg in 2019 (DA-BAR, 2022). Among the different onion varieties, bulb onions, particularly the Bermuda type, dominate production, comprising 98.6% of total output (PSA, 2023). Given the economic significance of onions and the challenges posed by price seasonality, this study analyzes the seasonal fluctuations of red onion prices through time series analysis, offering insights that can inform policy recommendations to stabilize prices and support local farmers.

Methodology

The study analyzed the seasonal price fluctuations of red onions in the Philippines by selecting major supply and demand areas based on production volume and market significance. Supply areas were identified using 2020 production data from the Philippine Statistics Authority (PSA), with Ilocos, Central Luzon, and MIMAROPA representing the primary onion-producing regions. The National Capital Region (NCR) was included as a key demand center, primarily sourcing onions from Central Luzon, Ilocos, and MIMAROPA. Additional demand areas were strategically chosen based on their roles in onion distribution: Region 4A serves as an entry point for onions from Region 4B, Region 7 (Central Visayas) functions as a major trading hub, particularly in Cebu, and Region 10 (Northern Mindanao) serves as both a trading center and a gateway for onions entering Mindanao.

Secondary data on the retail prices of red onions from 2012 to 2024 were obtained from the PSA. The degree of seasonal price variation was analyzed using a general time series model, applying the multiplicative model of the ratio-to-moving-average equation (Eq. 1):

$$P_t = T_t \times S_t \times C_t \times I_t \quad (1)$$

where P_t = red onion price at time t; T_t = trend; S_t = seasonal variation; C_t = cyclical variation; I_t = irregular movement

Results and Discussion

Volume of Production, Area, Prices, and Importation

Figure 1 illustrates the volume of bulb onion production in major growing areas in the Philippines from 2012 to 2023, based on data from the Philippine Statistics Authority (PSA, 2024). Over this period, onion production in the country exhibited an increasing trend, with an average growth rate of 13.06 percent. Central Luzon and MIMAROPA emerged as the top onion-producing regions, with production in these areas growing at 13.50 percent and 21.30 percent, respectively. Nueva Ecija, the leading onion-producing province in both Central Luzon and the entire country, recorded a production growth rate of 12.91 percent. Meanwhile, Occidental Mindoro in MIMAROPA and Pangasinan in the Ilocos Region also contributed significantly to the country's total bulb onion output.

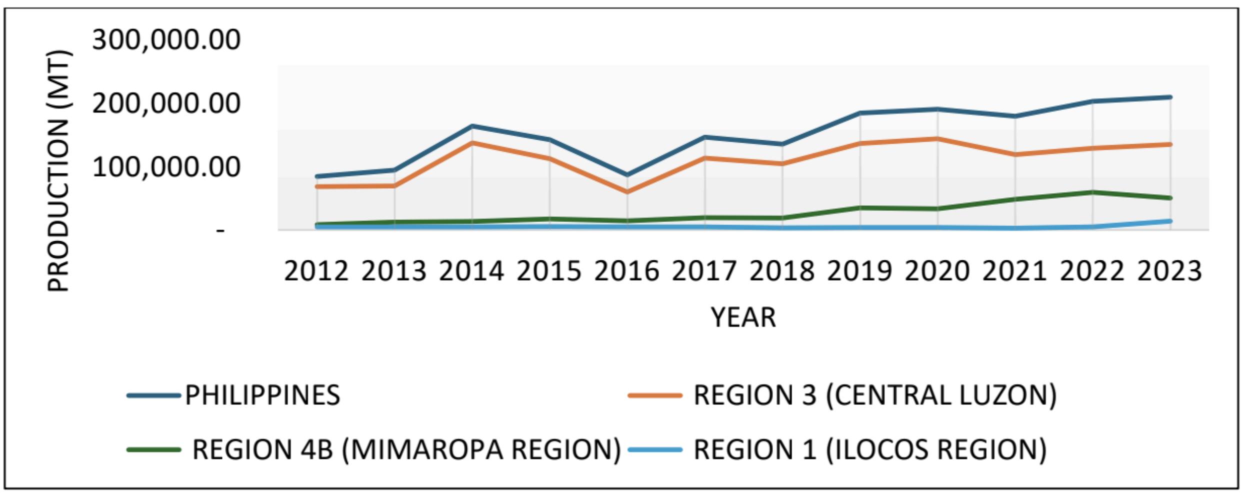

Figure 1. Volume of bulb onion production in major growing areas in the Philippines, 2012-2023 (Source: PSA, 2024)

As shown in Figure 2, the total area planted with bulb onions expanded at an average annual rate of 5.87 percent, with notable increases in Central Luzon (4.69%), MIMAROPA (12.45%), and the Ilocos Region (14.62%). This expansion in cultivated area contributed to the overall growth of onion production. However, as Figure 3 indicates, prices remained highly volatile, with pronounced peaks and troughs observed across farmgate, wholesale, and retail levels. The synchronized movements of these prices suggest external market influences which is possibly linked to supply shortages and importation policies.

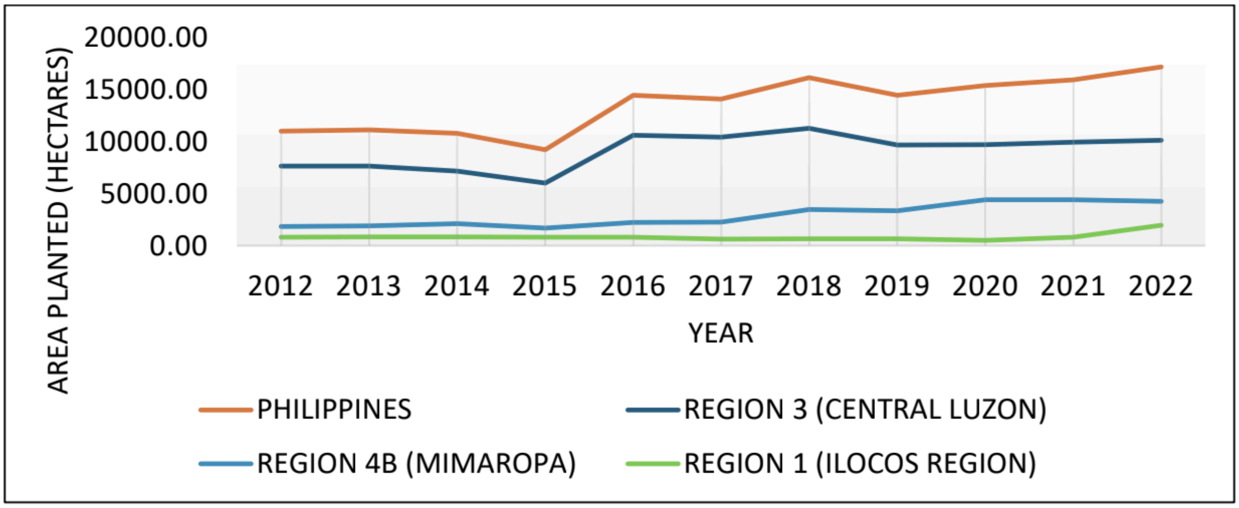

Figure 2. Area planted with bulb onion in major growing areas in the Philippines, 2012-2022 (Source: PSA, 2024)

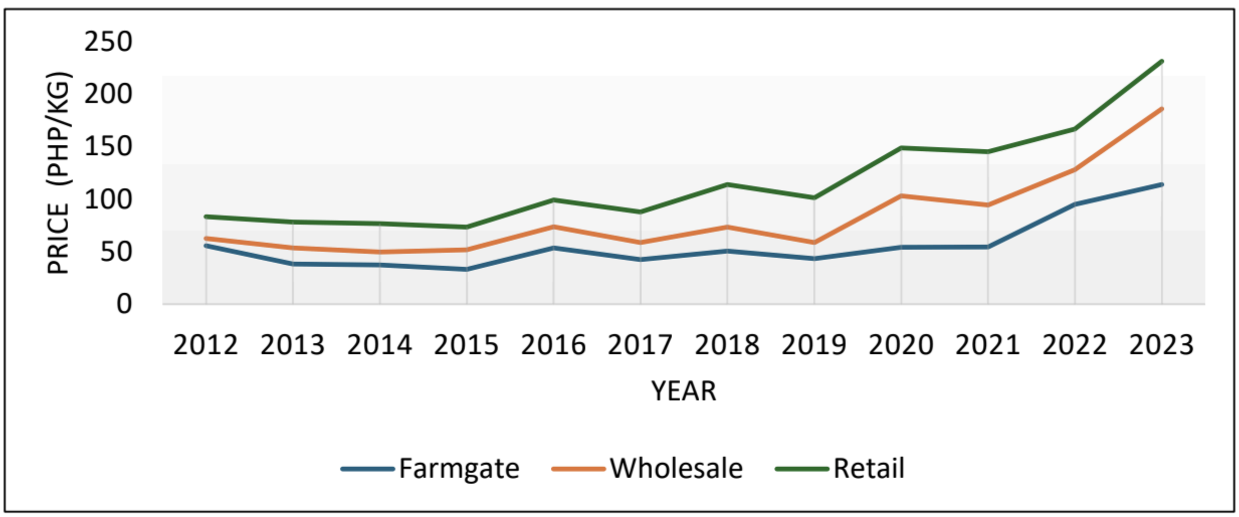

Figure 3. Trend in prices of red onion in the Philippines, 2012-2023 (Source: PSA, 2024)

Figure 4 provides further context by illustrating the trends in onion importation, which fluctuated significantly in response to production levels, market demand, and policy shifts. Import volumes declined sharply in 2013 and 2017 but surged in 2014, 2015, and 2016 due to supply shortages. A notable drop in 2019 was followed by a partial recovery in 2020. By 2023, import volumes had increased again, reflecting government interventions aimed at addressing supply deficits and stabilizing prices.

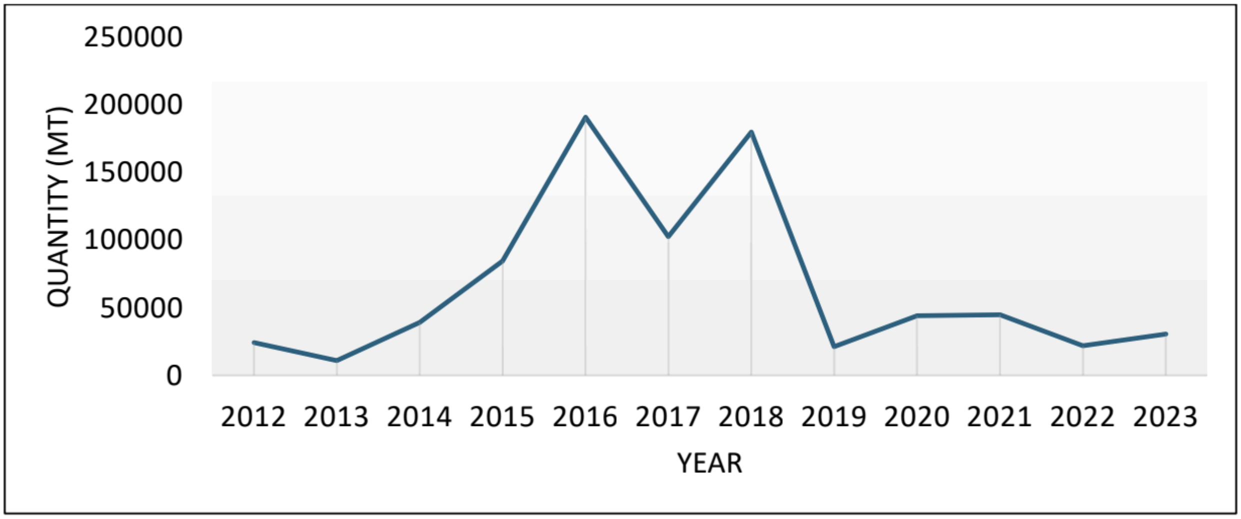

Figure 4. Trends in importation, Philippines, 2012-2023
(Source: UN Comtrade, 2024)

Seasonal Price Variation

Table 1 presents the seasonal price variation in the retail price of red onions in the Philippines from July 2012 to October 2024. Prices tend to be lowest from March to June, with the lowest seasonal price index typically observed in June, signaling the end of the harvest season. Conversely, the highest price indices occur from December to January, aligning with the holiday season and the start of the next harvest cycle.

Seasonal price indices below 1.00 indicate that prices during those months fall below the average price, with the gap reflecting the difference between 100 and the computed index value. Among the major onion-producing regions—Central Luzon, Ilocos Region, and MIMAROPA—the lowest seasonal price indices ranged from 0.7675 to 0.8180. Nationally, the highest seasonal price index of 126.73% suggests that retail prices in January were 26.73% higher than the average monthly price, while the lowest seasonal price index of 86.54% in May indicates that prices were 15.46% lower than the average.

These trends reflect the strong influence of seasonal production patterns on onion prices. September to December typically experiences price surges due to the planting season, with a subsequent decline from December to June as the harvest progresses.

Table 1. Seasonal price variation in the retail price of red onion in the Philippines, July 2012 to October 2024

Month	Philippines	Supply Centers			Demand Centers			
		Region 3	Region 1	Region 4B	NCR	Region 4A	Region 7	Region 10
January	126.73	135.77	132.76	130.68	127.70	130.68	134.38	118.89
February	111.16	109.13	111.01	111.96	106.37	111.96	118.29	111.77
March	95.46	84.51	87.43	94.16	90.06	94.16	99.26	98.64
April	87.56	76.75	79.97	84.13	85.50	84.13	87.33	92.34
May	86.54	79.43	81.85	81.80	87.13	81.80	84.39	92.38
June	88.83	85.86	86.01	84.05	87.98	84.05	83.13	95.64
July	90.38	89.50	89.03	87.86	90.02	87.86	87.05	94.61
August	93.23	95.43	94.91	93.75	93.55	93.75	89.49	94.85
September	96.35	100.98	100.20	98.51	96.97	98.51	92.90	94.57
October	100.74	107.68	107.20	103.47	102.10	103.47	99.43	96.95
November	105.62	111.17	110.20	107.75	108.88	107.75	104.48	99.96
December	117.40	123.76	119.44	121.90	123.74	121.90	119.86	109.38

Source: PSA, 2024

Irregular Price Movements

Figure 5a illustrates the irregular components of red onion retail prices in key supply centers from 2012 to 2024. Irregular price movements are characterized by fluctuations without a discernible pattern, often triggered by unforeseen events such as natural disasters, supply chain disruptions, or abrupt policy changes.

A negative quadratic trend was observed in these irregular price movements, indicating an inverse or indirect relationship over time. Despite this downward trend, a significant price spike occurred in early 2023, particularly in January, when retail prices reached their peak due to supply shortages and market disruptions. This was followed by a price decline and stabilization starting in February 2023, largely driven by increased onion importation as a government intervention.

Historical data from The Philippine Star (2024) reveals that during the 2022 holiday season, red onion prices surged above ₱700 per kilo, worsening supply shortages and causing white onions to become scarce. In response, the government imported large quantities of onions in 2023 to avoid another crisis. However, this caused farmgate prices to drop significantly to ₱10 per kilo, while retail prices remained high at ₱90–₱140, raising concerns about profiteering in the supply chain.

The impact of these price fluctuations has been particularly severe for farmers. Reports suggest that the same entities responsible for the 2022 onion crisis played a role in the 2023 price slump, reflecting systemic exploitation. Farmers continue to struggle due to inadequate infrastructure, particularly the lack of cold storage facilities, which forces them to sell their produce at low prices during peak harvest periods. Moreover, as younger generations increasingly abandon farming, the industry faces long-term sustainability challenges.

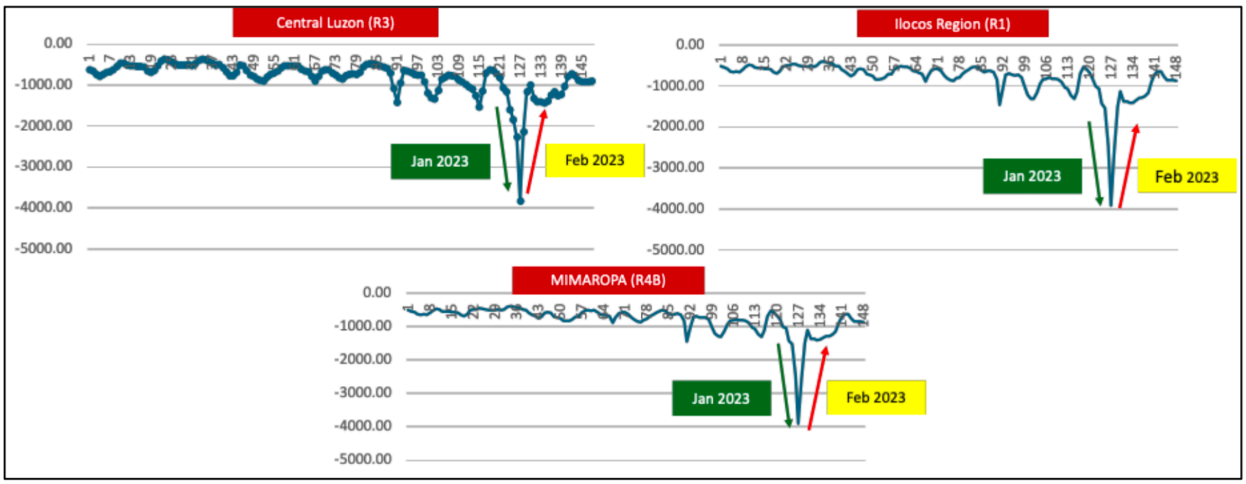

Figure 5a. Irregular components of red onion retail prices, supply centers, 2012-2024
(Source: PSA, 2024)

Figure 5b illustrates the irregular components of red onion retail prices in major demand centers from 2012 to 2024, highlighting significant price fluctuations caused by unforeseen market disruptions. The National Capital Region (NCR) experienced a peak in prices in December 2022, primarily driven by heightened holiday demand and constrained supply. In contrast, regions such as CALABARZON, Central Visayas, and Northern Mindanao saw their price peaks in January 2023, likely reflecting the delayed effects of supply shortages and logistical bottlenecks. Price stabilization varied across regions, with NCR and CALABARZON showing signs of recovery by February 2023, following government interventions such as increased imports and improved distribution. Central Visayas and Northern Mindanao followed suit, stabilizing by March 2023, reflecting regional differences in response times to market corrections and policy adjustments.

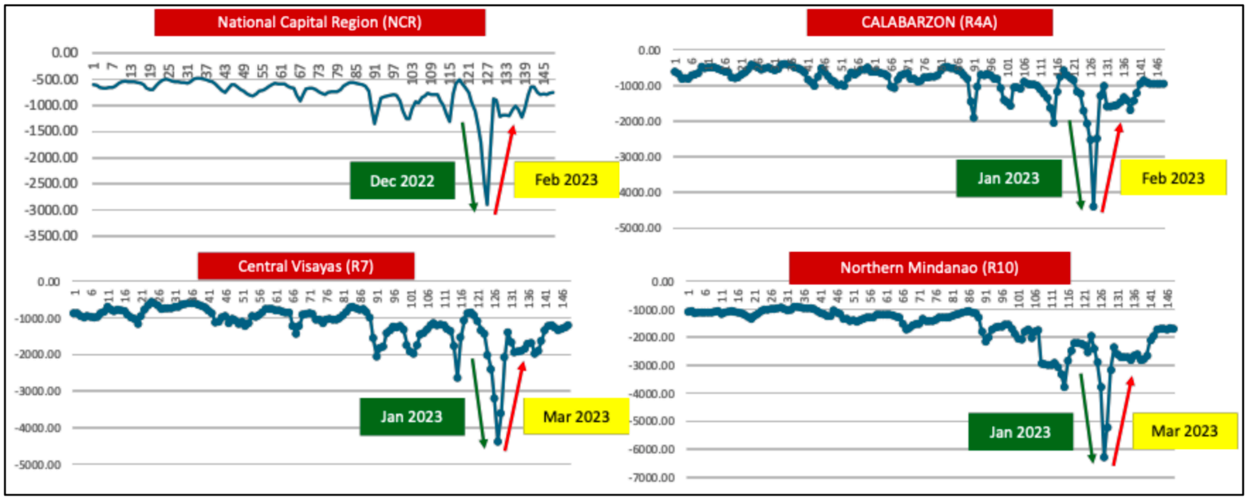

Figure 5b. Irregular components of red onion retail prices, demand centers, 2012-2024 (Source: PSA, 2024)

Conclusion and Recommendations

Onions, especially red onions, are essential in the Philippines due to their culinary and medicinal value, yet the onion market faces significant challenges, including seasonal price fluctuations and irregular price movements driven by production patterns, policy changes, and market inefficiencies. While production in regions like Central Luzon and MIMAROPA has been increasing, price volatility remains a persistent issue, affecting both farmers and consumers. Although import policies have mitigated shortages, they have not stabilized prices or protected small-scale farmers from exploitation. Regional disparities in price stabilization, with some areas recovering faster than others, highlight the need for region-specific interventions to improve market resilience. Farmers suffer financially from income instability due to low prices during surplus and high prices during shortages. While imports temporarily stabilize prices, the local market remains vulnerable to external shocks and policy changes. To address these issues, it is recommended to strengthen cold storage and post-harvest infrastructure to reduce losses, improve transparency in the supply chain to combat profiteering, develop regional trading centers to streamline distribution, and align importation policies with domestic production cycles to prevent oversupply during peak harvests. These measures will help stabilize the onion market, ensuring fair prices for both farmers and consumers.

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